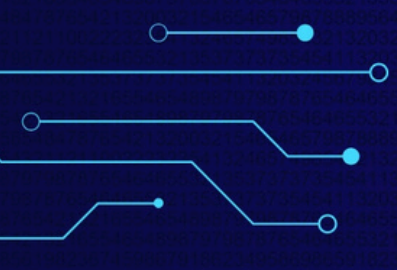




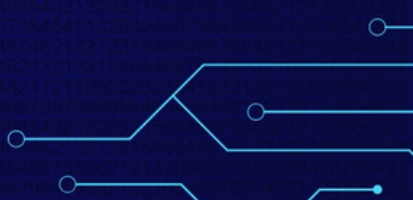
ADVANCE CLOUD SECURITY INTERVIEW QUESTIONS

**20 Expert Questions & Answers for SOC L2,
L3 & Cloud Security Professionals**



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ADVANCE CLOUD SECURITY INTERVIEW QUESTIONS

31. How would you investigate a suspected cloud account compromise?

Ideal Answer:

A suspected cloud account compromise should be treated as a high-priority incident because attackers can quickly access or modify cloud resources.

I would begin by:

- Identifying the affected account.
- Reviewing recent login activity.
- Checking login locations, devices, and IP addresses.
- Reviewing API activity.
- Checking whether MFA was enabled or bypassed.
- Identifying newly created users, roles, or access keys.

In AWS, I would review **CloudTrail**, while in Azure I would analyze **Azure Activity Logs** and **Microsoft Entra ID (Azure AD) Sign-in Logs**.

If compromise is confirmed, I would:

- Disable or lock the account.
- Revoke active sessions.
- Rotate passwords and access keys.
- Remove unauthorized changes.
- Continue monitoring for persistence.

32. What are the most common cloud attack vectors?

Ideal Answer:

Cloud attacks often occur because of misconfigurations or identity-related weaknesses rather than vulnerabilities in the cloud provider.

Common attack vectors include:

- Stolen credentials
- Publicly exposed storage buckets
- Weak IAM policies
- Misconfigured Security Groups or NSGs
- Compromised API keys
- Unpatched virtual machines

- Insecure APIs
- Phishing attacks
- Insider threats
- Excessive permissions

Understanding these attack vectors helps organizations prioritize cloud security controls.

33. What is Cloud Threat Hunting?

Ideal Answer:

Cloud Threat Hunting is the proactive process of searching cloud environments for malicious activity that has not yet triggered security alerts.

Threat hunters analyze:

- Authentication logs
- API calls
- Identity activity
- Resource modifications
- Network traffic
- Storage access
- Cloud workload behavior

Unlike traditional monitoring, cloud threat hunting assumes an attacker may already have access and actively searches for evidence of compromise.

34. How would you detect suspicious API activity in a cloud environment?

Ideal Answer:

Cloud services rely heavily on APIs, making API monitoring critical.

I would look for:

- Unusual API calls
- API calls from unfamiliar locations
- Privilege escalation requests
- Resource deletion
- IAM policy modifications
- New access key creation
- High-frequency API requests

CloudTrail in AWS and Azure Activity Logs provide visibility into API activity.

Suspicious API behavior may indicate compromised credentials or attacker activity.

35. What is the principle of Zero Trust in Cloud Security?

Ideal Answer:

Zero Trust is a security model based on the principle of "**Never Trust, Always Verify.**"

Instead of automatically trusting users inside the network, every request is continuously verified.

Zero Trust includes:

- Strong identity verification
- Least privilege access
- Continuous authentication
- Device verification
- Network segmentation
- Continuous monitoring

Zero Trust is particularly important in cloud environments where users and resources are distributed.

36. How would you investigate a publicly exposed storage bucket?

Ideal Answer:

Public storage buckets are one of the most common causes of cloud data exposure.

I would:

- Verify whether public access is intentional.
- Identify stored data.
- Review bucket permissions.
- Analyze access logs.
- Determine whether unauthorized access occurred.
- Review recent configuration changes.

If sensitive information is exposed, I would immediately restrict access and investigate potential data exposure.

37. Explain Cloud Security Posture Management (CSPM) and Cloud Workload Protection Platform (CWPP).

Ideal Answer:

Both are cloud security solutions but serve different purposes.

CSPM focuses on identifying cloud misconfigurations and compliance issues.

Examples include:

- Public storage buckets
- Weak IAM policies
- Missing encryption
- Disabled logging

CWPP focuses on protecting workloads such as:

- Virtual machines
- Containers
- Kubernetes clusters
- Serverless workloads

Organizations often deploy both CSPM and CWPP together for comprehensive cloud protection.

38. What is Container Security?

Ideal Answer:

Container Security involves protecting containerized applications throughout their lifecycle.

Security considerations include:

- Secure container images
- Vulnerability scanning
- Image signing
- Runtime monitoring
- Least privilege containers
- Network isolation
- Secrets management

Containers should be continuously monitored because vulnerabilities can exist in both applications and container images.

39. What are Kubernetes security risks?

Ideal Answer:

Common Kubernetes security risks include:

- Misconfigured RBAC permissions
- Exposed Kubernetes API server
- Privileged containers

- Insecure secrets storage
- Vulnerable container images
- Excessive permissions
- Weak network policies

Organizations should secure both the Kubernetes control plane and worker nodes while continuously monitoring cluster activity.

40. What is Serverless Security?

Ideal Answer:

Serverless Security focuses on securing functions that execute without dedicated servers.

Examples include:

- AWS Lambda
- Azure Functions

Important security considerations include:

- IAM permissions
- Secure code
- Dependency management
- Event validation
- Logging
- Monitoring

Although infrastructure management is handled by the cloud provider, customers remain responsible for securing application logic and permissions.

41. How would you investigate privilege escalation in a cloud environment?

Ideal Answer:

Privilege escalation occurs when an attacker gains higher permissions than originally assigned.

I would investigate:

- IAM policy changes
- Role assignments
- New administrative accounts
- API activity
- Privileged login events
- Access key creation

I would compare current permissions against historical configurations to identify unauthorized privilege changes.

42. What is the importance of cloud logging during Incident Response?

Ideal Answer:

Cloud logs provide a detailed record of attacker activity.

During an investigation, logs help identify:

- Initial access
- Authentication attempts
- API activity
- Configuration changes
- Resource creation
- Data access
- Privilege escalation

Without logging, determining the scope and timeline of a cloud incident becomes significantly more difficult.

43. What is Identity Federation in Cloud Security?

Ideal Answer:

Identity Federation allows users to authenticate using an external identity provider instead of maintaining separate cloud credentials.

Common identity providers include:

- Microsoft Entra ID
- Active Directory
- Okta
- Google Workspace

Benefits include:

- Centralized authentication
- Single Sign-On (SSO)
- Simplified user management
- Improved security

Federation reduces password management complexity and supports enterprise identity management.

44. How do attackers maintain persistence in cloud environments?**Ideal Answer:**

Cloud persistence techniques differ from traditional endpoint persistence.

Attackers may:

- Create new IAM users
- Generate new API keys
- Modify IAM roles
- Create scheduled automation
- Deploy malicious serverless functions
- Configure external identity trust
- Modify storage permissions

During investigations, analysts should review identity changes and administrative activity for signs of persistence.

45. What is Cloud Forensics?**Ideal Answer:**

Cloud Forensics is the process of collecting, preserving, and analyzing evidence from cloud environments after a security incident.

Evidence sources include:

- CloudTrail
- Azure Activity Logs
- Identity logs
- Virtual machine snapshots
- Storage access logs
- Network flow logs

Cloud forensics helps reconstruct attacker activity while maintaining evidence integrity.

46. What challenges are unique to Cloud Incident Response?**Ideal Answer:**

Cloud environments introduce several challenges, including:

- Limited access to underlying infrastructure
- Shared responsibility model
- Dynamic resource creation

- Multi-region deployments
- Short-lived workloads
- Identity-based attacks
- Multi-cloud visibility

Incident responders must understand cloud-native logging and security services to investigate effectively.

47. How would you secure cloud workloads?

Ideal Answer:

Cloud workloads should be secured by:

- Applying least privilege
- Enabling MFA
- Encrypting data
- Regular patching
- Vulnerability scanning
- Runtime monitoring
- Logging and alerting
- Network segmentation
- Endpoint protection
- Continuous compliance assessments

Security should be integrated throughout the workload lifecycle.

48. What is a Multi-Cloud Environment, and what are its security challenges?

Ideal Answer:

A Multi-Cloud Environment uses services from more than one cloud provider, such as AWS and Azure.

Advantages include:

- Vendor flexibility
- High availability
- Disaster recovery

Security challenges include:

- Different IAM models
- Multiple security tools

- Inconsistent policies
- Complex monitoring
- Compliance management

Organizations should standardize security controls across all cloud platforms.

49. How does Cloud Security differ from Traditional On-Premises Security?

Ideal Answer:

Traditional environments focus heavily on securing physical infrastructure.

Cloud environments emphasize:

- Identity security
- Configuration management
- API security
- Shared responsibility
- Elastic infrastructure
- Automation

Cloud security requires continuous monitoring because resources can be created or modified rapidly.

50. Describe a complete Cloud Incident Response process.

Ideal Answer:

A structured cloud incident response process includes:

1. Detect suspicious activity through alerts or monitoring.
2. Validate whether the activity represents a security incident.
3. Determine the affected accounts, resources, and regions.
4. Collect evidence from cloud logs and telemetry.
5. Contain the incident by disabling accounts, revoking sessions, or restricting access.
6. Eradicate the threat by removing malicious users, keys, or configurations.
7. Recover affected resources and restore normal operations.
8. Continuously monitor for signs of reinfection or persistence.
9. Conduct a root cause analysis to understand how the incident occurred.
10. Document findings, update security controls, and improve detection rules and response procedures.

Following a structured process helps minimize business impact while improving the organization's overall cloud security posture.